

PULSEGRID • Complete UML Architectural Specification

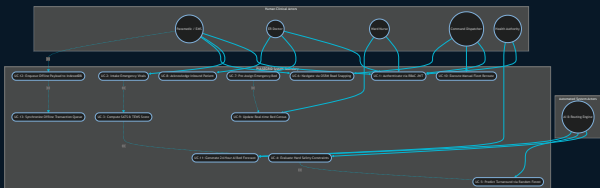
Intelligent Emergency Routing & Bed Management System • Zero-API-Cost Telemetry & Geospatial Triage Platform

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1. UML Use Case Diagram

SYSTEM BOUNDARY & ACTORS

Defines the functional requirements and interaction boundaries between human clinical actors (Paramedics, ER Doctors, Ward Nurses, Dispatchers, Health Authorities) and the automated AI recommendation engine.

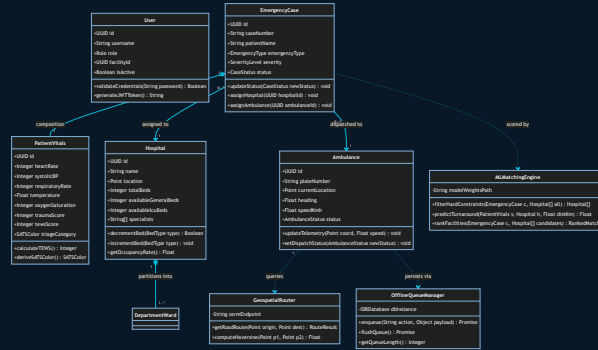


Examiner Defense Point: Note the strict <<include>> dependency chaining: Vital intake automatically computes SATS/TEWS, which enforces hard clinical constraints before executing machine learning scoring. Offline persistence acts as a decoupled <<extend>> event.

2. UML Class Diagram (Domain Model)

OBJECT-ORIENTED ARCHITECTURE

Specifies domain entities, typed attributes, visibility rules (+ public, - private), operations, and entity multiplicities across the PULSEGRID core engine.



Examiner Defense Point: The composition relationship between EmergencyCase and PatientVitals ensures that vitals cannot exist independently of a clinical case, maintaining medical audit compliance under Ghana's Act 843.

3. Sequence Diagram: Emergency Intake & AI Triage

SYNCHRONOUS DATA FLOW

Chronological execution of paramedic vital input, SATS/TEWS classification, PostGIS spatial radius query (60km), and sub-millisecond Random Forest prediction.



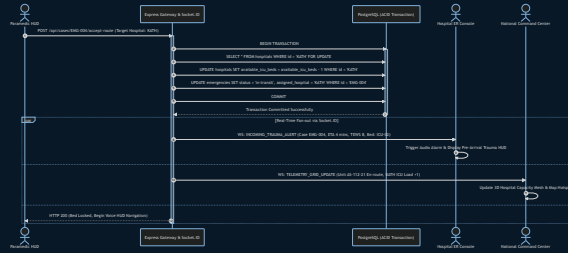
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Examiner Defense Point: Emphasize the microsecond ML inference latency (11.1 μ s). Because inference is sub-millisecond, the bottleneck is purely network I/O, allowing instantaneous updates on moving paramedic cockpits.

4. Sequence Diagram: Pre-Arrival Bed Lock & Fan-out

REAL-TIME TELEMETRY
BROADCAST

Atomic database transactions lock critical beds before arrival, triggering simultaneous WebSocket broadcasts to emergency physicians and city dispatchers.

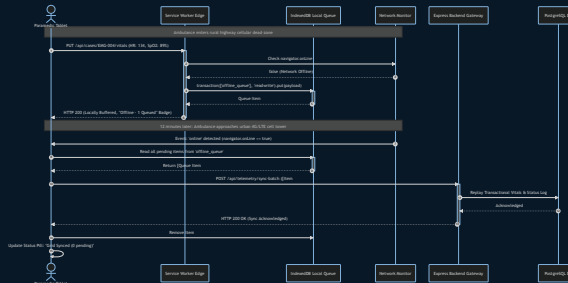


Examiner Defense Point: The `SELECT ... FOR UPDATE` statement guarantees ACID serializability, preventing race conditions where two ambulances heading to the same hospital simultaneously claim the final ICU bed.

5. Sequence Diagram: Dead-Zone Offline Sync

EDGE FAULT TOLERANCE

Demonstrates the two-tier edge resilience strategy when an ambulance traverses a cellular dead-zone on national highway corridors.

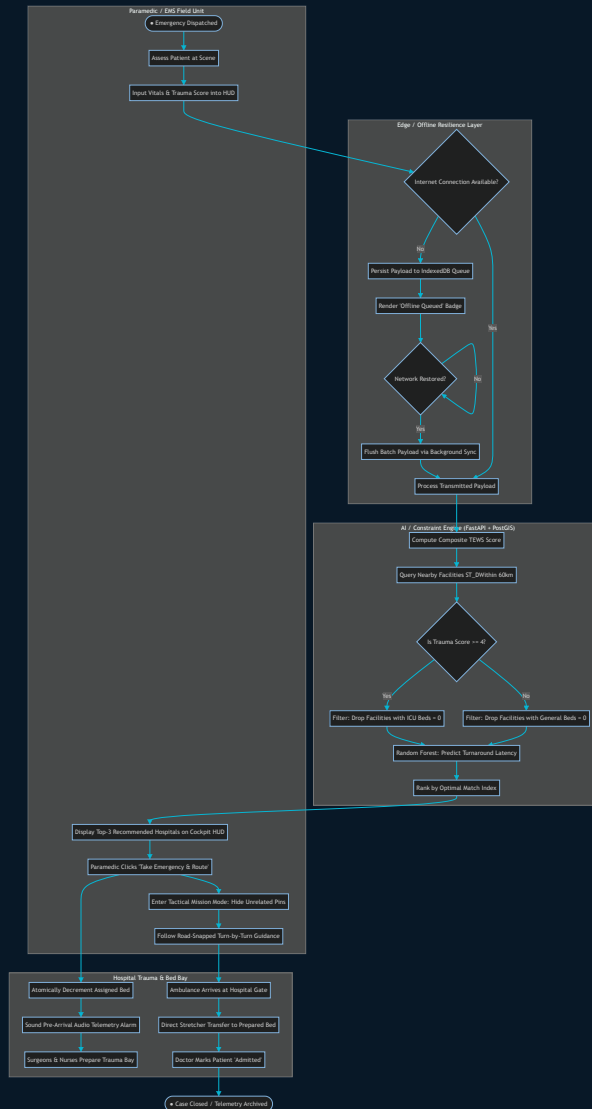


Examiner Defense Point: Proves full resilience against rural Ghana telecom instability. The system never throws unhandled network rejection errors; payloads are queued locally in browser-native IndexedDB and flushed automatically upon reconnection.

6. UML Activity Diagram (Clinical Swimlanes)

OPERATIONAL CONTROL FLOW

Parallel swimlane activity flow coordinating paramedic field actions, offline edge buffering, AI constraint execution, and trauma hospital admission.

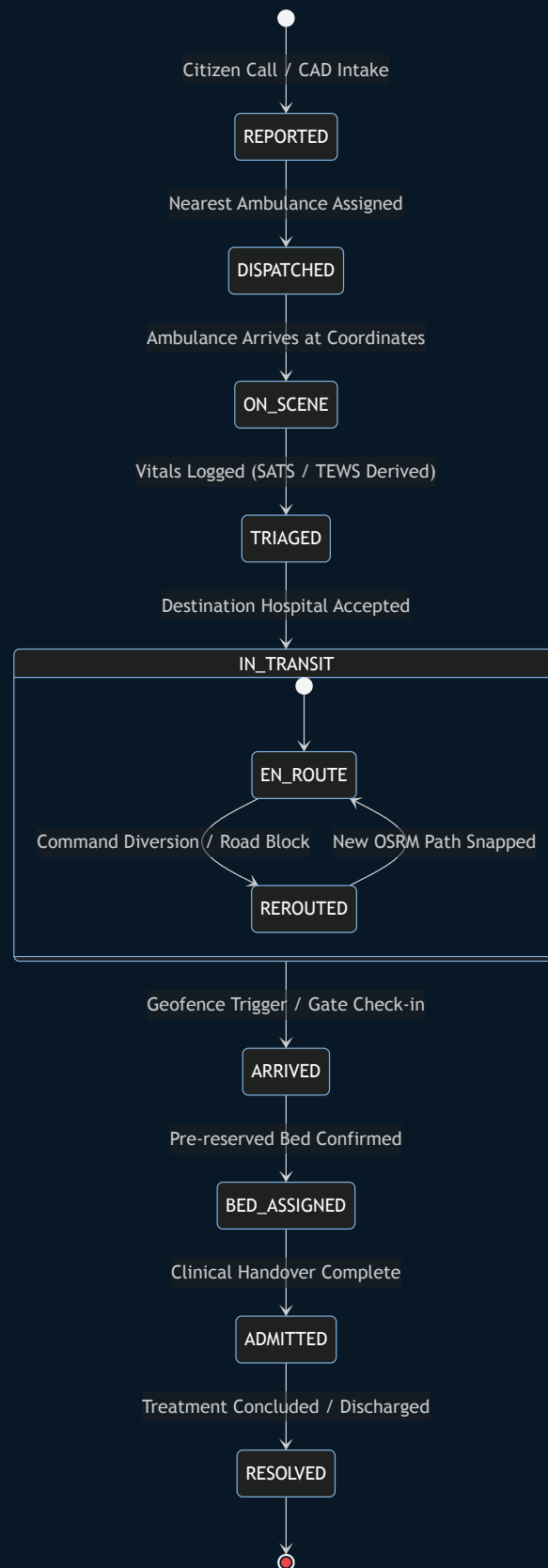


Examiner Defense Point: Trace the decision branch `Is Trauma Score >= 4?`. This is where medical guidelines override naive distance-based routing, preventing severe trauma patients from being diverted to clinics with zero ICU capabilities.

7. UML State Machine Diagram

ENTITY LIFECYCLE MANAGEMENT

Statechart modeling all valid state transitions and nested sub-states for emergency incidents and fleet telemetry units.



Examiner Defense Point: The composite IN_TRANSIT state allows dynamic re-routing (e.g. if the patient's condition deteriorates en-route, or the command center triggers a mass casualty protocol)

without resetting the incident lifecycle.

8. UML Deployment Diagram

HARDWARE NODES, NETWORK & CLOUD TOPOLOGY

Architectural deployment mapping physical edge mobile tablets, Vercel CDN nodes, backend Docker containers, PostgreSQL/PostGIS databases, and communication protocols.



Examiner Defense Point: Emphasize microservice isolation: The compute-heavy spatial graph (OSRM) and the machine learning model (FastAPI) are containerized separately from the Node.js API Gateway, preventing routing CPU spikes from degrading live WebSocket latency.